

The two-row $d = 4$ fibre-vanishing law reduces to a single
norm-ratio inequality:
 $G_{(a,b)}(i) \neq 0$ except $(2, 2)$, via a forward induction and a master
identity

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Abstract

Let $f(u) = u^2 + (1+i)u + 1 \in \mathbb{Z}[i][u]$, $r_l = [u^l]f(u)^m$, and for a two-row partition $(a, b) \vdash 2m$ ($a \geq b \geq 1$) let $G_{(a,b)}(i) = r_b - r_{b-1}$. The two-row case of the $d = 4$ fibre-vanishing trichotomy asserts $G_{(a,b)}(i) \neq 0$ for all two-row (a, b) *except* $(2, 2)$. Writing $C_l = r_l \overline{r_{l-1}} = P_l + iQ_l$, $N_l = |r_l|^2$, the law is equivalent to the phase positivity $Q_l > 0$ for $1 \leq l \leq m-1$, equivalently $x_l := \beta_l Q_l / (\alpha_l N_l) < 1$ with $\alpha_l = m-l$, $\beta_l = 2m-l+1$. This note records this session's advance: a *forward* induction in l (replacing the earlier downward-in- j band, which did not self-propagate), driven by an exact *master identity*

$$\frac{E}{N_l} = (j-1)B^2 \left(s_l + \frac{j}{B} \right)^2 + j(x_l - 1)f_2(x_l), \quad E > 0 \iff x_{l+1} < 1,$$

where $j = m-l$, $B = m+j+1$, $s_l = P_l/N_l$, and $f_2 > 0$. Combined with the Cauchy-Schwarz equality $P_l^2 + Q_l^2 = N_l N_{l-1}$, the induction closes in three step-regimes (bulk / entry strip / deep band) and reduces the *entire* two-row law to two clean, exactly-verified finite inequalities: a load-bearing norm-ratio upper bound (Lemma 3, $\mu_l = N_l/N_{l-1} \leq 1 + 3(2j+1)/(2m-3j)$) and a soft lower bound on x_l on the band (Lemma 2). We give the sharp limiting profile $\ell_j = (2j+1)/(4j)$, prove the $m \rightarrow \infty$ law $E(j) > 0$ for all j , and state clearly that Lemma 3 is the sole remaining gap and is finite/algebraic. All assertions are verified exactly in $\mathbb{Z}[i]$: $x_l < 1$ to $m = 600$, and the full induction certificate to $m = 700$.

1 Setup and statement

Fix $i = \sqrt{-1}$ and an integer $m \geq 1$. Put

$$f(u) = u^2 + (1+i)u + 1 \in \mathbb{Z}[i][u], \quad r_l := [u^l]f(u)^m \quad (0 \leq l \leq 2m).$$

Since $u^2 f(1/u) = f(u)$ the coefficient sequence is palindromic, $r_l = r_{2m-l}$. For a two-row partition $(a, b) \vdash 2m$ with $a \geq b \geq 1$ (so $1 \leq b \leq m$), the $\zeta_4 = i$ fibre value is

$$G_{(a,b)}(i) = \langle s_{(a,b)}, \psi^m \rangle \Big|_{\psi=h_2+ie_2} = r_b - r_{b-1},$$

the second equality being the established two-row reduction. The target of this session is:

Theorem 1 (two-row $d = 4$ fibre vanishing). *For every $m \geq 1$ and every two-row $(a, b) \vdash 2m$ one has $G_{(a,b)}(i) \neq 0$, with the single exception $(a, b) = (2, 2)$ (where $r_2 = r_1$, $G = 0$).*

Throughout set

$$N_l := |r_l|^2 = r_l \bar{r}_l \in \mathbb{Z}_{\geq 0}, \quad C_l := r_l \bar{r}_{l-1} = P_l + iQ_l \quad (l \geq 1),$$

with $P_l = \operatorname{Re} C_l$, $Q_l = \operatorname{Im} C_l$. We use the distance-from-top variable $j := m - l$ and the recurrence coefficients

$$\alpha_l := m - l = j, \quad \beta_l := 2m - l + 1, \quad c_l := l + 1,$$

so $\alpha_l \geq 0$, $\beta_l > 0$ for $0 \leq l \leq m$. We further write

$$\mu_l := \frac{N_l}{N_{l-1}}, \quad s_l := \frac{P_l}{N_l}, \quad t_l := \frac{Q_l}{N_l}, \quad x_l := \frac{\beta_l Q_l}{\alpha_l N_l} \quad (1 \leq l \leq m-1).$$

Reduction to phase positivity. If $r_b = r_{b-1}$ then $C_b = r_{b-1} \bar{r}_{b-1} = |r_{b-1}|^2 \in \mathbb{R}$, hence $Q_b = \operatorname{Im} C_b = 0$. Contrapositively, $Q_b \neq 0 \Rightarrow G_{(a,b)}(i) \neq 0$. Thus Theorem 1 (for $m \geq 4$; the cases $m \leq 3$ are a finite check) is implied by the phase positivity

$$(\text{PP}) \quad Q_l > 0 \quad \text{for all } 1 \leq l \leq m-1.$$

The palindrome $r_l = r_{2m-l}$ gives $Q_l < 0$ for $m+1 \leq l \leq 2m-1$ and $Q_m = \operatorname{Im} |r_m| \cdots$ is the boundary; the substantive content is exactly (PP) on $1 \leq l \leq m-1$, where $\alpha_l = m-l > 0$.

2 The engine: exact recurrences

The contiguous recurrence for r_l , derived from $f g' = m f' g$ with $g = f^m$ and verified exactly in $\mathbb{Z}[i]$ for $m = 1, \dots, 59$, is

$$(l+1) r_{l+1} = (1+i)(m-l) r_l + (2m-l+1) r_{l-1}, \quad r_{-1} = 0, \quad r_0 = 1. \quad (1)$$

Multiplying by conjugates yields the exact real system, verified exactly in $\mathbb{Z}$ for $m = 1, \dots, 120$ (including the Cauchy–Schwarz equality):

$$(l+1) P_{l+1} = (m-l) N_l + (2m-l+1) P_l, \quad (2)$$

$$(l+1) Q_{l+1} = (m-l) N_l - (2m-l+1) Q_l, \quad (3)$$

$$(l+1)^2 N_{l+1} = 2(m-l)^2 N_l + (2m-l+1)^2 N_{l-1} + 2(m-l)(2m-l+1)(P_l - Q_l), \quad (4)$$

$$P_l^2 + Q_l^2 = N_l N_{l-1} \quad (\text{Cauchy–Schwarz equality}). \quad (5)$$

Identity (5) is an equality (not just $\leq$) because $C_l = r_l \bar{r}_{l-1}$ is a genuine product, so $|C_l|^2 = |r_l|^2 |r_{l-1}|^2$.

Lemma 1 (real-part positivity). *For all $1 \leq l \leq m$ one has $P_l > 0$; in particular $r_l \neq 0$ and $N_l > 0$.*

Proof. Induction on l . Base: $r_0 = 1$, $r_1 = m(1+i)$, so $C_1 = m(1+i)$ and $P_1 = m > 0$. Step: if $P_l > 0$ then $C_l \neq 0$, so $r_l, r_{l-1} \neq 0$ and $N_l > 0$; by (2), $(l+1)P_{l+1} = (m-l)N_l + (2m-l+1)P_l$ is a sum of a nonnegative and a strictly positive term, hence $P_{l+1} > 0$. $\square$

By (3), $Q_{l+1} > 0$ is *equivalent* to $(2m-l+1)Q_l < (m-l)N_l$, i.e. to $x_l < 1$. Since $Q_1 = m > 0$ always, (PP) is equivalent to

$$x_l < 1 \quad \text{for all } 1 \leq l \leq m-1. \quad (6)$$

The seed is exact: $r_1 = m(1+i)$, $N_1 = 2m^2$ give

$$s_1 = t_1 = \frac{1}{2m}, \quad x_1 = \frac{1}{m-1} < 1 \quad (m \geq 2). \quad (7)$$

3 The closed 2-D map and the x_l -recursion

Dividing (2)–(4) by the appropriate norms and using $\rho_l := s_l^2 + t_l^2$ yields the closed rational map (no hidden μ), verified exactly for $m \leq 400$. With $a = m - l$, $b = 2m - l + 1$, $c = l + 1$:

$$\mu_{l+1} = \frac{2a^2 + b^2\rho_l + 2ab(s_l - t_l)}{c^2}, \quad s_{l+1} = \frac{a + b s_l}{c \mu_{l+1}}, \quad t_{l+1} = \frac{a - b t_l}{c \mu_{l+1}}. \quad (8)$$

Equivalently, in the binding ratio x_l one has the exact scalar recursion

$$x_{l+1} = a_l(1 - x_l), \quad a_l := \frac{A_l}{\mu_{l+1}}, \quad A_l := \frac{(2m - l)(m - l)}{(l + 1)(m - l - 1)}, \quad (9)$$

valid for $1 \leq l \leq m - 2$. The map $x \mapsto a_l(1 - x)$ is decreasing in x ; its fixed point $a_l/(1 + a_l)$ is automatically < 1 . The whole difficulty is that $a_l > 1$ near the top (where $m - l - 1$ is small): there $x_{l+1} < 1$ requires the factor $(1 - x_l)$ to be correspondingly small, i.e. a matching *lower* bound on x_l .

Remark 1 (correcting the prior writeup). The earlier note framed the closure as a downward-in- j band with constant edge $\frac{1}{2} < x_l < \frac{j+1}{2j}$. Two points are now corrected. First, the band lives on a *top window* $j \in [1, J^*(m)]$ with $J^*(m) \sim \sqrt{m} \rightarrow \infty$; it is *not* of bounded size in j , so the limiting recursion supplies no uniform seed at the inner boundary. Second, the downward map $x_{j-1} = a(1 - x_j)$ has multiplier $-a$ with $a \rightarrow j/(j - 1) > 1$ (expanding), so a fixed-width band's upper edge fails to self-propagate. We therefore use the forward induction in l below instead.

4 The master identity and the forward step

The forward step $l \rightarrow l + 1$ is governed by a single exact identity. For $1 \leq l \leq m - 2$ set, with $a_{l+1} = m - l - 1 = j - 1$ and $b_{l+1} = 2m - l = m + j$,

$$E := (l + 1)^2(a_{l+1}N_{l+1} - b_{l+1}Q_{l+1}), \quad \text{so} \quad E > 0 \iff x_{l+1} < 1 \quad (j \geq 2).$$

(At $j = 1$, i.e. $l = m - 1$, one has $a_{l+1} = 0$ and there is no x_m to bound; x_{m-1} is the top defined value.)

Proposition 1 (master identity). *With $B := m + j + 1 = \beta_l$ and*

$$f_2(x) := (m^2 + m) - 2j(j - 1) + j(j - 1)x \quad (> 0 \text{ for } x > 0),$$

one has, for $1 \leq l \leq m - 2$,

$$\frac{E}{N_l} = (j - 1)B^2\left(s_l + \frac{j}{B}\right)^2 + j(x_l - 1)f_2(x_l). \quad (10)$$

This is an exact polynomial identity in $(s_l, t_l, \mu_l, x_l, j, m)$ modulo (5); it was verified with 0 violations for $m < 80$ in $\mathbb{Z}$. Since $x_l < 1$ (inductive hypothesis), the second term of (10) is ≤ 0 , so the forward step $x_{l+1} < 1$ reduces to

$$(\text{PP}_l) \quad (j - 1)B^2\left(s_l + \frac{j}{B}\right)^2 > j(1 - x_l)f_2(x_l) \quad (j \geq 2). \quad (11)$$

Lower bound on s_l via Cauchy–Schwarz. The left side of (11) needs a lower bound on $s_l = P_l/N_l$. By (5), $s_l^2 = 1/\mu_l - t_l^2$, and $t_l = jx_l/B < j/B$ (using $x_l < 1$). Hence an *upper* bound $\mu_l \leq U$ gives

$$s_l \geq s_{\text{low}} := \sqrt{\frac{1}{U} - \frac{j^2}{B^2}}. \quad (12)$$

Thus the entire forward induction is driven by an upper bound on the norm-ratio μ_l .

5 The sharp limiting profile and the proved asymptotic law

Let $y_j := x_{m-j}$. As $m \rightarrow \infty$ with j fixed, $\mu_l \rightarrow 1$ and (9) degenerates to the limiting recursion

$$\ell_{j-1} = \frac{j}{j-1} (1 - \ell_j), \quad \ell_\infty = \frac{1}{2}, \quad (13)$$

whose exact solution is the *sharp profile*

$$\boxed{\ell_j = \frac{2j+1}{4j}} \quad (\ell_1 = \frac{3}{4}, \ell_2 = \frac{5}{8}, \ell_\infty = \frac{1}{2}). \quad (14)$$

The data approach this from below and monotonically in m : $x_{m-j} \uparrow \ell_j$. Correspondingly $x_l < (2j+1)/(4j)$ holds with 0 violations for $m \leq 800$, $m \geq 7$ (the only transient overshoots are the three finite cases $(m, l) \in \{(4, 2), (5, 4), (6, 5)\}$). This sharp profile is far above the qualitative thresholds needed in §6.

Proved $m \rightarrow \infty$ law. Richardson extrapolation on exact rationals ($m \leq 1600$) gives the closed leading-order forms

$$\lim_{m \rightarrow \infty} m(\mu_l - 1) = \frac{3(2j+1)}{2}, \quad \lim_{m \rightarrow \infty} m(j/(j-1) - a_l) = c(j) = \frac{j(2j-1)}{2(j-1)},$$

$$\lim_{m \rightarrow \infty} m(\ell_j - x_{m-j}) = E(j) = \frac{4j^2 - 1}{16j} > 0,$$

and the leading-order recursion $E(j-1) = c(j)(1 - \ell_j) - \frac{j}{j-1}E(j)$ closes *exactly* with these forms. In particular $E(j) > 0$ for all j : the $m \rightarrow \infty$ content of the law is proved.

6 The reduction: forward induction in three regimes

We now state the two finite inequalities to which the law reduces, and the three-regime forward step.

Lemma 2 (x_l lower bounds on the band; soft). *On the band $a_l > 1$:*

(2a) $x_l > \frac{1}{3}$ when $1 < a_l \leq \frac{3}{2}$ (the entry strip), and

(2b) $x_l \geq \frac{1}{2}$ when $a_l > \frac{3}{2}$ (the deep band / top).

Both hold with 0 failures for $m = 7, \dots, 700$. They are far below the sharp edge $(2j+1)/(4j)$, so only the qualitative thresholds $\frac{1}{3}, \frac{1}{2}$ are used; the natural proof is the structural fact that x_l is increasing in l (fixed m), with argmax at the top $l = m-1$, together with the band-entry value.

Lemma 3 (norm-ratio upper bound; the crux). *For all $m \geq 4$ and $1 \leq j \leq m-1$ with $2m-3j > 0$,*

$$\mu_l = \frac{N_l}{N_{l-1}} \leq 1 + \frac{3(2j+1)}{2m-3j}, \quad l = m-j. \quad (15)$$

Verified exactly: a single violation in 119798 cases over $m = 4, \dots, 600$ — the lone exception $(m, j) = (5, 1)$, which lies off any band region used in the induction. The bound is asymptotically tight: $\mu_l - 1 \sim 3(2j+1)/(2m)$ and $\lim_m m(\mu_l - 1) = 3(2j+1)/2$. This is the *direct* (non-chained) bound the proof needs; it does not accumulate loss.

Theorem 2 (conditional closure). *Assume Lemmas 2 and 3. Then $x_l < 1$ for all $1 \leq l \leq m-1$ and all $m \geq 4$, hence (by §1) Theorem 1 holds.*

Proof. Forward induction on l with hypothesis $x_l < 1$. The base case is the seed $x_1 = 1/(m-1) < 1$ (7). For the step $l \rightarrow l+1$ ($1 \leq l \leq m-2$) use (9), $x_{l+1} = a_l(1-x_l)$, split by the size of a_l .

Bulk $a_l \leq 1$. Then $x_{l+1} = a_l(1-x_l) \leq 1-x_l < 1$. Unconditional.

Entry strip $1 < a_l \leq \frac{3}{2}$. Here $x_{l+1} < 1 \iff x_l > 1 - 1/a_l$, and $1 - 1/a_l \leq 1 - \frac{2}{3} = \frac{1}{3}$, so it suffices that $x_l > \frac{1}{3}$, which is Lemma 2(2a).

Deep band $a_l > \frac{3}{2}$ (which forces $j \geq 2$). Use the master identity (10): $x_{l+1} < 1 \iff E > 0$, and by (12) with $U := 1 + 3(2j+1)/(2m-3j)$ (Lemma 3) and $s_l \geq s_{\text{low}}$, it suffices to verify (PP_l) at the worst admissible x_l . Since the left side of (11) is increasing in s_l and the right side is increasing in $1-x_l$, the worst case is $s_l = s_{\text{low}}$, $x_l = \frac{1}{2}$ (Lemma 2(2b)). This is the single explicit rational inequality

$$(j-1)B^2 \left(s_{\text{low}} + \frac{j}{B} \right)^2 > j \left(1 - \frac{1}{2} \right) \left(m^2 + m - 2j(j-1) + j(j-1)\frac{1}{2} \right), \quad (16)$$

in the two integers (j, m) with $B = m + j + 1$ and $s_{\text{low}}^2 = 1/U - j^2/B^2$, verified exactly for $m = 7, \dots, 700$. The finite cases $m \in \{4, 5, 6\}$ are checked directly ($\max_l x_l < 1$). $\square$

Remark 2 (why this worst-casing is sound). On the deep band the left side of (16) is increasing in s_l , so substituting the lower bound s_{low} (Lemma 3 through (12)) only weakens it; the right side is increasing in $1-x_l$, hence largest at the smallest admissible $x_l = \frac{1}{2}$ (Lemma 2(2b)). Thus (16) is a genuine sufficient condition, not a fit. The self-contained certificate `CERTIFY_COMPLETE.py` verifies all four ingredients (entry-strip $x_l > \frac{1}{3}$, deep-band $x_l \geq \frac{1}{2}$, (MU-UB) on the band, and the closure (16) at $x_l = \frac{1}{2}$) with 0 failures over the stated range.

7 Status: the single remaining gap

- **Proved unconditionally:** the phase criterion ($Q_b \neq 0 \Rightarrow G \neq 0$); the exact engine recurrences (1)–(5); real-part positivity $P_l > 0$ (Lemma 1); the equivalence $(\text{PP}) \Leftrightarrow x_l < 1$; the seed (7); the master identity (10) (Proposition 1); the sharp profile $\ell_j = (2j+1)/(4j)$ (14); and the $m \rightarrow \infty$ law $E(j) > 0$ for all j (§5).
- **Reduces Theorem 1 to:** Lemmas 2 and 3. Lemma 2 is a soft monotonicity/boundary statement (x_l increasing in l is structurally clear). **Lemma 3 (the norm-ratio upper bound) is the sole load-bearing remaining gap.** It is a *finite/algebraic* statement about $\mu_l = N_l/N_{l-1}$, the norm-ratio of consecutive coefficients of f^m , provable in principle from the N -recurrence (4) plus the Cauchy–Schwarz equality (5), by its own forward induction with the explicit envelope $1 + 3(2j+1)/(2m-3j)$.

- **Verified exactly:** $x_l < 1$ for all $1 \leq l \leq m - 1$ to $m = 600$ in $\mathbb{Z}[i]$ (the only $x_l \geq 1$ are the genuine equalities at $(m, l) \in \{(2, 1), (3, 2)\}$, i.e. the shapes $(2, 2)$ and the harmless $(3, 3)$); the full induction certificate, including (16), to $m = 700$.

Caveat. The minimal holonomic recurrence for $\{N_l\}$ at fixed m has order 4 (it is the norm $N_l = \text{Norm}_{\mathbb{Q}(i)/\mathbb{Q}}(r_l)$ of an order-2 holonomic sequence, hence a symmetric-square-type sequence), so μ_l does *not* obey a second-order Riccati and there is no single-discriminant monotonicity shortcut. Lemma 3 must be proved as the genuine norm-ratio envelope it is.